a) 2:1 b) 1:2:1 c) 3:1 d) **1:1:1:1**

12. Pure line breed refers to (AIIMS 2002, AIIMS 2007)
a) Heterozygosity only
b) Heterozygosity and linkage
c) **Homozygosity only**
d) Homozygosity and self assortment

13. How many different types of gametes can be formed by F_1 progeny, resulting from the following cross AABBCC x aabbcc (AIIMS 2004)
a) 3 b) **8** c) 27 d) 64

14. Which of the following conditions represents a case of co-dominant genes? (AIIMS 2009)
a) A gene expresses itself, suppressing the phenotypic effect of its alleles
b) Genes that are similar in phenotypic effect when present separately, but when together interact to produce a different trait
c) Alleles both of which interact to produce a trait which may or may not resemble either of the parental type
d) Alleles, each of which produces an independent effect in a heterozygous condition.

15. If 'A' represents the dominant gene and 'a' represents its recessive allele, which of the following would be most likely result in the first generation off spring when Aa is crossed with aa? (AIIMS 2016)
a) All will exhibit dominant phenotype
b) All will exhibit recessive phenotype
c) **Dominant and recessive phenotypes will be 50% each**
d) Dominant phenotype will be 75%

16. In *Pisum Sativum*, there are 14 chromosomes. How many types of homologous pairs can be prepared? (JIPMER 2010)
a) 14 **b) 7** c) 2^{14} d) 2^{10}

17. The year 1900 AD is highly significant for geneticists due to (JIPMER 2013)
a) Discovery of genes
b) Principle of linkage
c) Chromosomal theory of heredity
d) **Rediscovery of Mendelism**

18. The phenotypic ratio of trihybrid cross in F_2 generation is (JIPMER 2016)
a) **27:9:9:9:3:3:3:1** b) 9:3:3:1
c) 1:4:6:4:1 d) 27:9:3:3:9:1:2:1

19. In a mutational event when adenine is replaced by guanine, it is the case of (AIPMT 2004)
a) Frameshift mutatin b) Transcription
c) **Transition** d) Transversion

20. Mutations can be induced with (AIPMT 2011)
a) **Gamma radiations** b) Infrared radiations
c) IAA d) Ethylene

21. The mechanism that causes a gene to move from one linkage group to another is called (AIPMT 2015, NEET (Phase – II) 2016)
a) **Translocation** b) Crossing over
c) Inversion d) Duplication

22. A point mutation comprising the substitution of a purine by pyrimidine is called (AIIMS 2002)
a) Transition b) Translocation
c) Deletion d) **Transversion**

23. Frameshift mutation occurs when (AIPMT 2008)
a) Base is substituted
b) **base is deleted or added**
c) Anticodons are absent
d) None of these

24. The distance between two genes in a chromosome is measured in cross-over units which represent (AIIMS 2008)
a) Ratio of crossing over between them
b) **Percentage of crossing over between them**
c) Number of crossing over between them
d) None of these

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_200-238 CH 10.indd 212 3/14/2020 12:18:02 PM